E-commerce Customer Behavior Analysis

By

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Batch Code:

Business Problem

- 🛒 **Understanding Customer Behavior in E-Commerce**
- E-commerce companies want to understand how customers interact with their websites and how their behavior affects purchasing decisions.
- Without proper analysis, it becomes difficult for companies to improve marketing strategies, enhance user experience, and increase sales.
- This project analyzes customer behavior using data analysis and machine learning to generate useful insights for better business decision-making.

Business Objectives 🛒

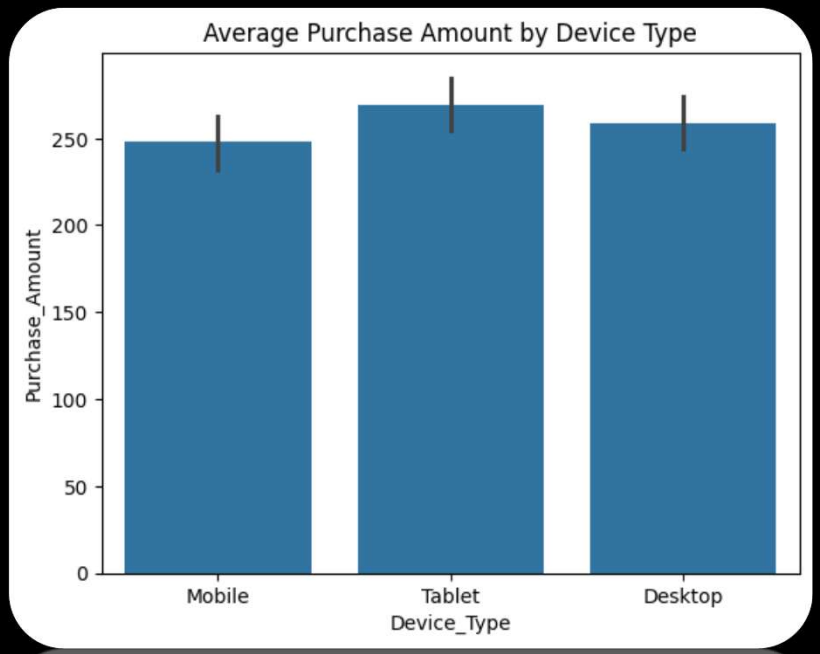
- To understand how customers interact with the e-commerce website by analyzing their age, device type, number of pages viewed, and session duration during their visit.
- Analyse customer purchasing behavior and find out which factors affect the purchase amount and membership level of customers.

Dataset Description

- **E-commerce Customer Behavior Dataset**
- **Total Records:** 1000 Customer Sessions
- **Key Variables in the Dataset:-**

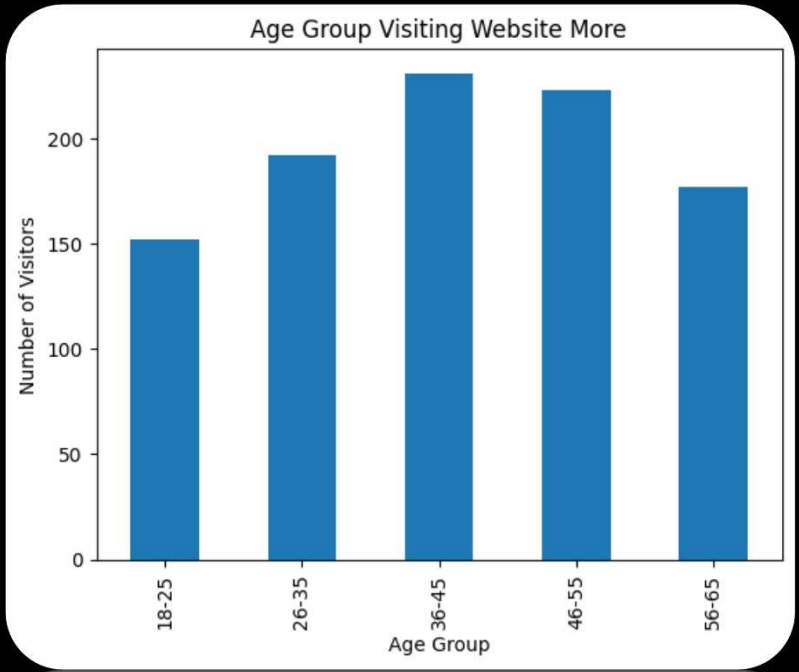
👤 **Customer_ID** – Unique identifier for each customer
🎂 **Age** – Age of the customer
📱 **Device_Type** – Device used (Mobile, Desktop, Tablet)
⏱️ **Session_Duration** – Time spent on the website
📄 **Pages_Viewed** – Number of pages visited during the session
💰 **Purchase_Amount** – Amount spent by the customer
⭐ **Membership_Level** – Customer loyalty or membership tier

Which device type is mostly used by customers who make the highest purchase?



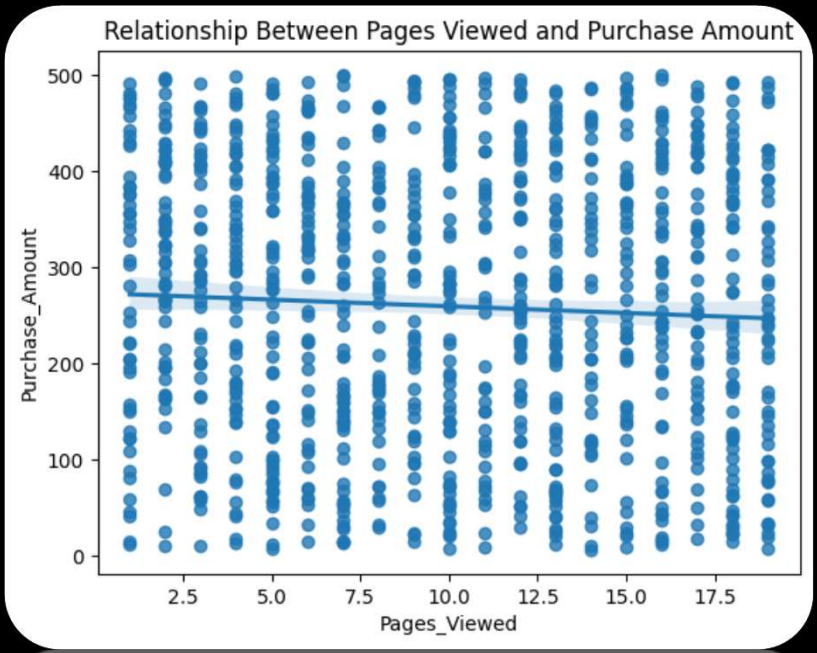
The chart shows the average purchase amount by device type.
Tablet users have the highest purchase amount compared to mobile and desktop users.
This indicates that device type may influence customer spending behavior.

Which age group of customers visit the website more?



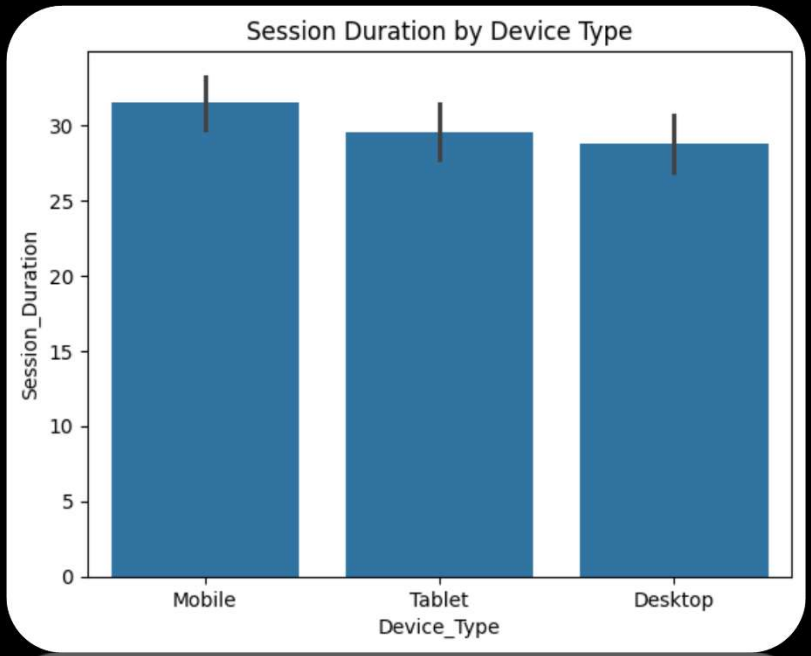
The chart shows website visits by different age groups.
Customers aged 36–45 visit the website the most.
This indicates that middle-aged customers are more active on the platform.

Customers who view few pages spend less or more money?



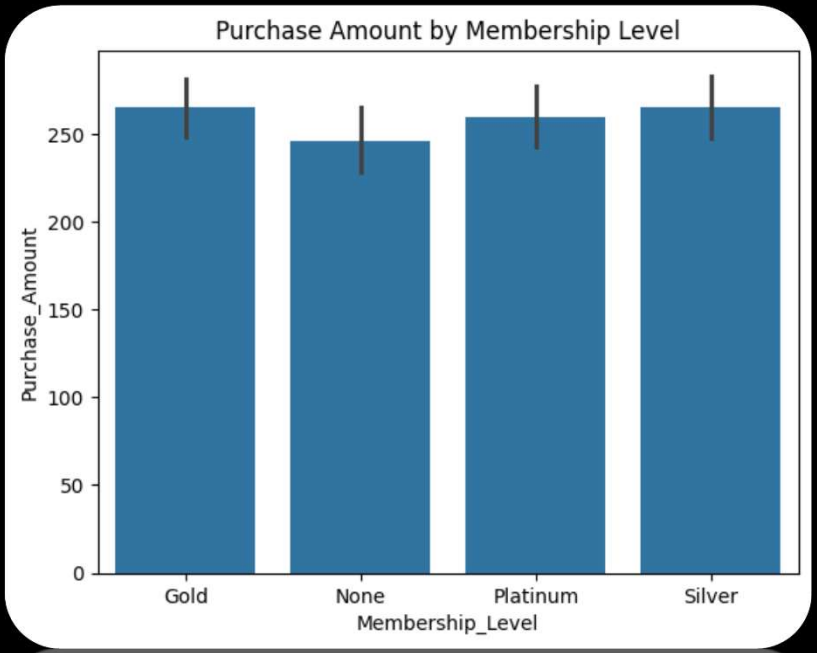
The scatter plot shows the relationship between pages viewed and purchase amount.
The trend line is almost flat, showing a weak relationship between them.
This means viewing more pages does not strongly affect how much customers spend.

Which device type users have the lowest session duration?



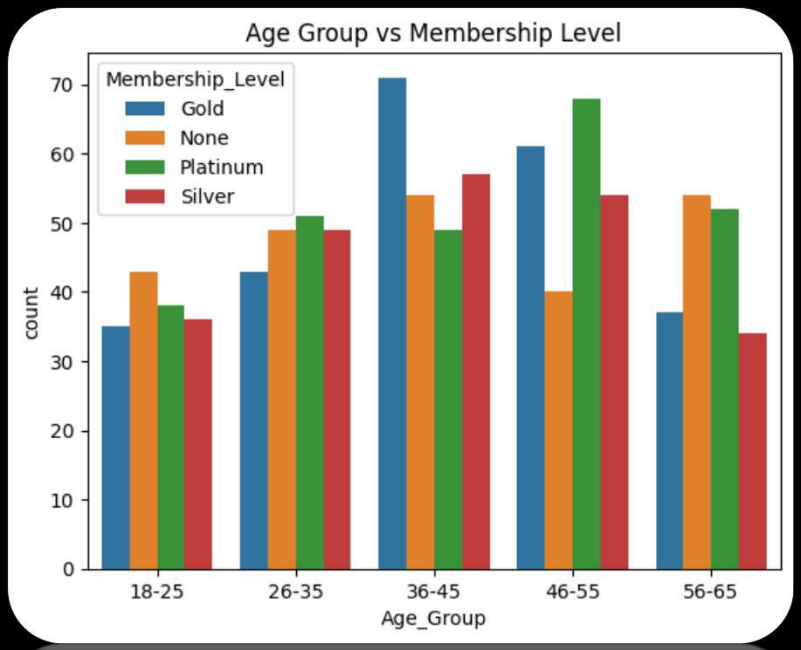
The chart shows average session duration by device type.
Desktop users have the lowest session duration compared to mobile and tablet users.
This means desktop users spend less time on the website.

Which membership level has the highest purchase amount?



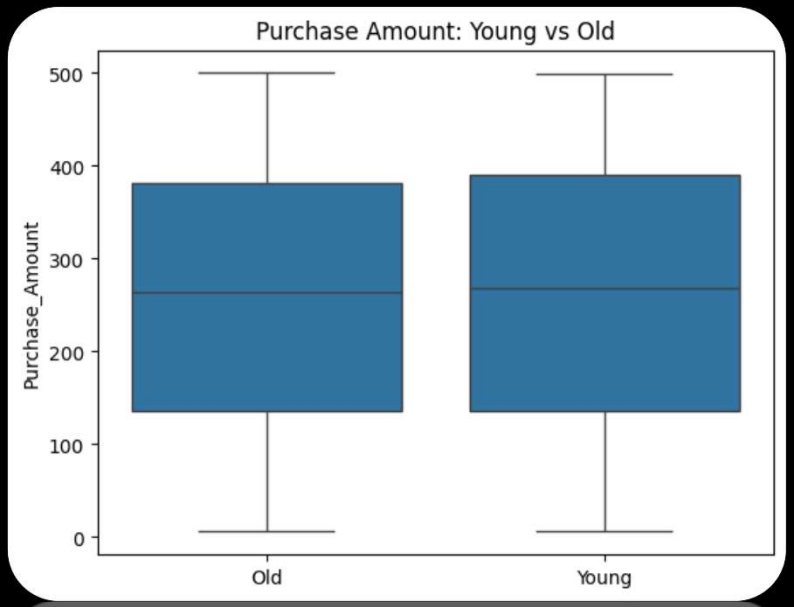
The chart shows purchase amount by membership level.
Silver members have the highest average purchase amount.
This indicates that membership level may influence customer spending.

Does age affect membership level?



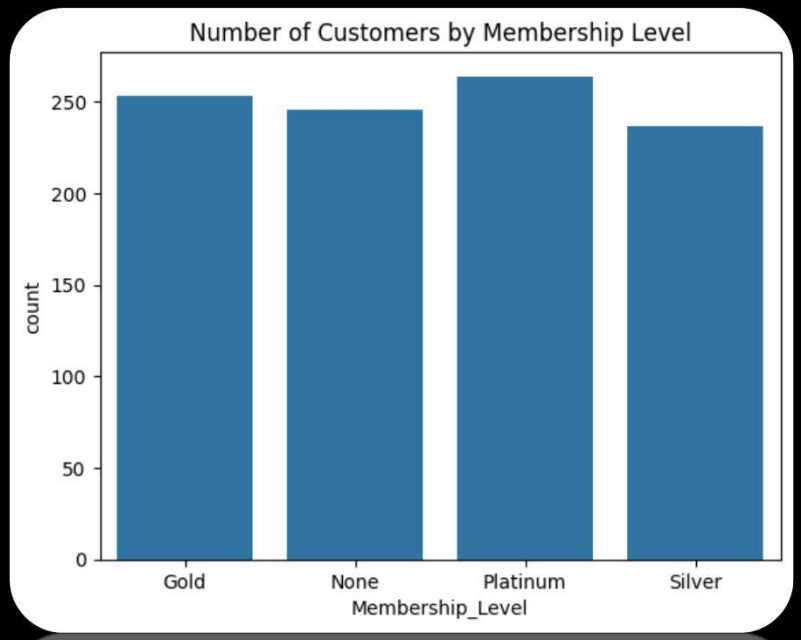
The chart shows the relationship between age group and membership level.
Different age groups have different membership distributions.
This suggests that age may influence the choice of membership level.

Difference in purchase amount between younger and older customers



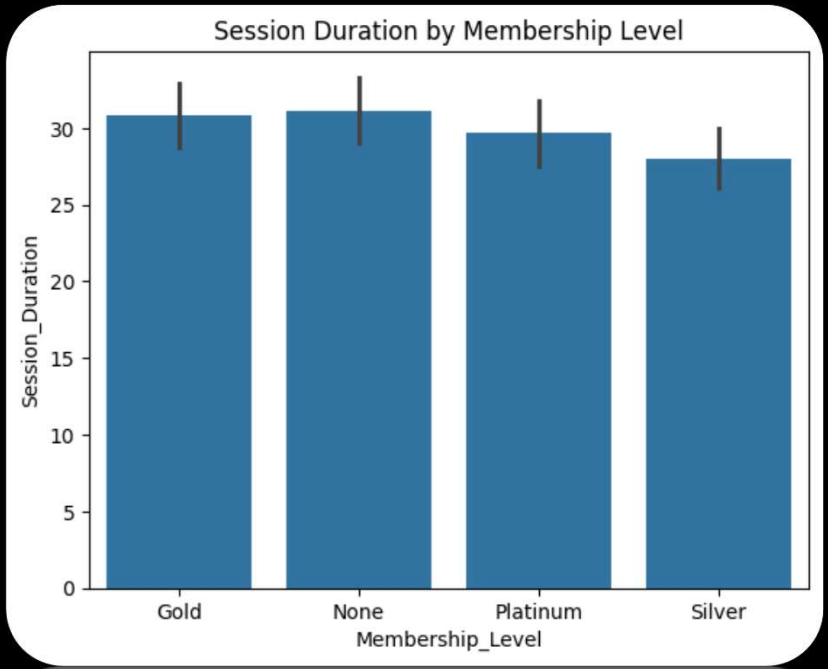
The box plot compares purchase amounts of younger and older customers.
Both groups show similar spending patterns.
This suggests that age does not strongly affect purchase amount.

Which membership level has the highest number of customers?



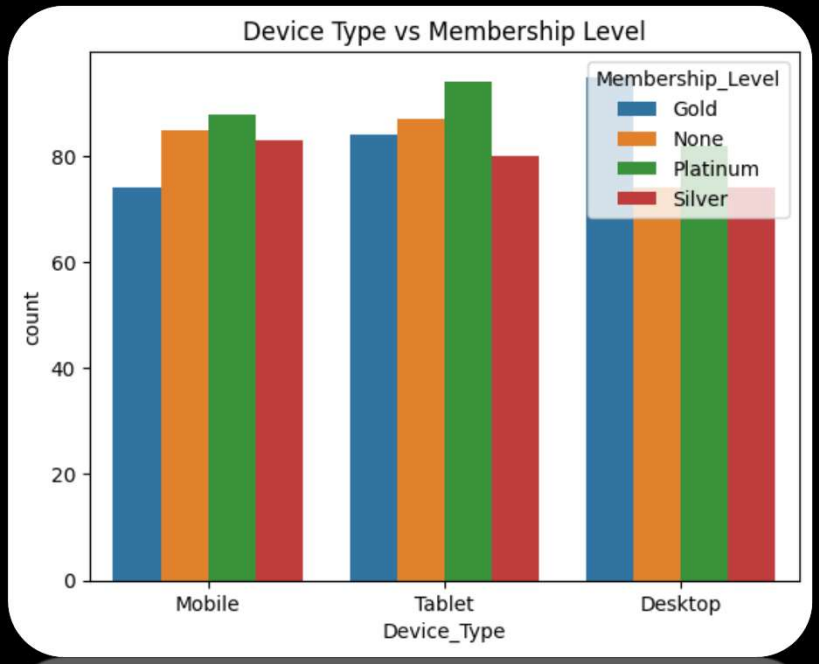
The chart shows the number of customers in each membership level.
Platinum membership has the highest number of customers.
This indicates that Platinum is the most popular membership level.

Which membership level has the highest session duration?



The chart shows session duration by membership level.
Customers with no membership have the highest session duration.
This means non-members spend slightly more time on the website.

Relationship between Device Type and Membership Level



The chart shows the relationship between device type and membership level.
Platinum membership appears most across all device types.
This indicates that many Platinum members use different devices to access the website.

Statistical Approach

- **Statistics helped transform raw customer data into meaningful business insights.**
- **Key Analyses Performed**
- 🔹 **Independent T-Test**
  Compared the average purchase amount between **young and old customers** to check if age influences spending behavior.
- **AFTER T-TEST :-** Young and old customers spend almost the same amount on the website.
- 🔹 **Chi-Square Test**
  Analyzed whether **device type and membership level** are related.
- **AFTER CHI-SQUARE TEST :-** The relationship is not significant because the membership distribution is similar across all device types

Machine Learning Approach

- 🤖 **Machine Learning techniques were applied to analyze customer behavior and generate insights.**
- 🔷 **Logistic Regression**
  Used to classify whether a customer is a member or non-member based on features such as age, session duration, pages viewed, purchase amount, device type, and engagement behavior. Feature engineering was applied by creating Engagement Score and Pages per Minute, and membership levels were simplified into binary classes to improve model performance. After these improvements, the model accuracy increased from 25% to 55%.
- 🔷 **K-Nearest Neighbors (KNN)**
  Used to classify customers as **member or non-member** based on features like age, session duration, pages viewed, purchase amount, device type, and engagement metrics. The optimal **K = 14** was selected, and the model achieved **72% accuracy**.

Summary

- **Key Work Done in the Project**
- 🔹 Performed **Exploratory Data Analysis (EDA)** to understand the dataset and customer behavior.
- 🔹 Applied **Statistical Analysis** such as t-test, and chi-square test to discover relationships in the data.
- 🔹 Built a **Logistic Regression model** to classify customers as member or non-member.
  🔹 Applied **K-Nearest Neighbors (KNN)** to classify customers as member or non-member based on their browsing and engagement behavior.

Conclusion

- The project analyzed **customer behavior on an e-commerce platform** using data analysis, statistical methods, and machine learning techniques.
- **Key Findings**
- 🔹 Customer features such as **age, session duration, pages viewed, device type, and purchase amount were analyzed to understand customer engagement and membership behavior.**
- 🔹 The **Logistic Regression model** was used to classify customers as **member or non-member**, and feature engineering helped improve the model accuracy.
- 🔹 The **K-Nearest Neighbors (KNN) model** further classified customers based on similar behavior patterns and achieved **72% accuracy**
- **Final Insight:-**
- The study demonstrates that **data analysis and machine learning can help businesses better understand customer behavior, identify patterns, and support data-driven decision making in e-commerce.**

Thank You